

# Firmware Anomaly Detection

(AI and Embedded based Model)

Project Name: Embedded Anomaly Guard

# Executive Summary

## Problem Statement / Objective:

If software manipulated or software is running additional malicious tasks, it required to be detected and cured.

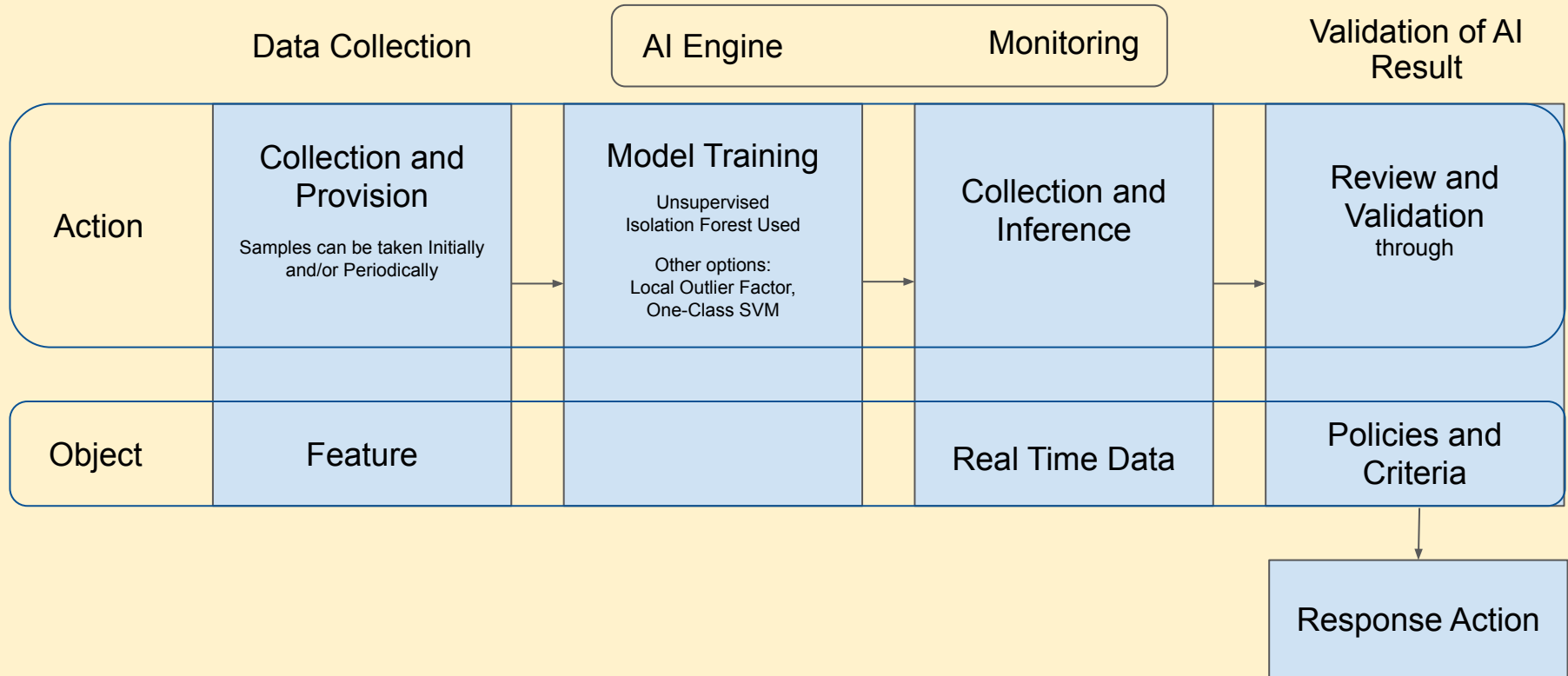
## Exploited Outcome Use Case:

- Change in Behavior of Application
- Unauthorized Data Collection / Modification

## Solution:

Understand the normal operational routine, use it as a reference and monitor the system to identify if any anomaly occurs.

# Architectural Model - AI based Solution for Application Processor



# Proof of Concept Execution - AI based

## Feature Collection

```
bonny-p@Nebula: ~/Documents/Projects/EAG_IDS
File Edit Tabs Help
bonny-p@Nebula:~/Documents/Projects/EAG_IDS $ python3 Data_Collector_EAG.py

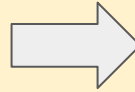
Embedded Anomaly Guard
=====
Data collector started.

Configuration:
Sampling interval : 5 seconds
Maximum records  : 10
CSV logging       : True
Averaging         : True

Parameters:
✓ cpu_percent
✓ ram_percent
✓ process_count

Press Ctrl+C to stop.

2026-09-18T21:31:24 | CPU: 0.2% | RAM: 67.5% | Processes: 213
2026-09-18T21:31:29 | CPU: 0.3% | RAM: 67.4% | Processes: 213
2026-09-18T21:31:34 | CPU: 0.0% | RAM: 67.3% | Processes: 213
2026-09-18T21:31:39 | CPU: 1.5% | RAM: 67.2% | Processes: 213
2026-09-18T21:31:44 | CPU: 0.3% | RAM: 67.2% | Processes: 213
2026-09-18T21:31:49 | CPU: 2.3% | RAM: 67.2% | Processes: 213
2026-09-18T21:31:54 | CPU: 0.0% | RAM: 67.2% | Processes: 212
- Average saved: system_average.csv
2026-09-18T21:31:59 | CPU: 0.2% | RAM: 67.1% | Processes: 212
2026-09-18T21:32:04 | CPU: 0.0% | RAM: 67.2% | Processes: 212
2026-09-18T21:32:09 | CPU: 0.2% | RAM: 67.1% | Processes: 212
2026-09-18T21:32:14 | CPU: 3.3% | RAM: 67.2% | Processes: 212
2026-09-18T21:32:19 | CPU: 0.5% | RAM: 67.2% | Processes: 212
```



## AI based self learned and generated Output - Detection

```
bonny-p@Nebula: ~/Documents/Projects/EAG_IDS
File Edit Tabs Help
bonny-p@Nebula:~/Documents/Projects/EAG_IDS $ python3 Monitor_EAG.py

Embedded Anomaly Guard
=====
Real-Time Monitor

AI features:
✓ cpu_percent
✓ ram_percent
✓ process_count

Loading training data: system_data.csv
Training samples: 10

Training Isolation Forest...
AI model trained.

Starting real-time monitoring...
Press Ctrl+C to stop.

2026-09-18 21:40:35 | CPU: 3.8% | RAM: 68.4% | Processes: 218 | NORMAL | Score: 0.1041
2026-09-18 21:40:40 | CPU: 5.3% | RAM: 68.3% | Processes: 216 | NORMAL | Score: 0.1035
2026-09-18 21:40:45 | CPU: 5.0% | RAM: 68.1% | Processes: 217 | NORMAL | Score: 0.1076
2026-09-18 21:40:50 | CPU: 5.5% | RAM: 68.0% | Processes: 217 | NORMAL | Score: 0.1076
2026-09-18 21:40:55 | CPU: 3.8% | RAM: 67.9% | Processes: 217 | NORMAL | Score: 0.1102
2026-09-18 21:41:00 | CPU: 37.6% | RAM: 65.4% | Processes: 217 | NORMAL | Score: 0.0337
2026-09-18 21:41:05 | CPU: 61.8% | RAM: 70.4% | Processes: 217 | ANOMALY | Score: -0.0352
2026-09-18 21:41:10 | CPU: 77.7% | RAM: 83.0% | Processes: 218 | ANOMALY | Score: -0.0679
2026-09-18 21:41:15 | CPU: 80.0% | RAM: 83.5% | Processes: 218 | ANOMALY | Score: -0.0679
2026-09-18 21:41:20 | CPU: 0.3% | RAM: 68.7% | Processes: 217 | NORMAL | Score: 0.0838
2026-09-18 21:41:25 | CPU: 1.0% | RAM: 68.3% | Processes: 215 | NORMAL | Score: 0.1218
2026-09-18 21:41:30 | CPU: 0.7% | RAM: 68.1% | Processes: 217 | NORMAL | Score: 0.1000
2026-09-18 21:41:35 | CPU: 1.5% | RAM: 68.0% | Processes: 217 | NORMAL | Score: 0.1060
2026-09-18 21:41:40 | CPU: 0.8% | RAM: 67.8% | Processes: 218 | NORMAL | Score: 0.1029
2026-09-18 21:41:45 | CPU: 1.5% | RAM: 67.7% | Processes: 219 | NORMAL | Score: 0.1089
```

Python based executable available @ [https://github.com/bonny-p/Embedded\\_Anomaly\\_Guard](https://github.com/bonny-p/Embedded_Anomaly_Guard)

Platform: Linux, Windows, MacOS

Dependencies: pip - psutils, pandas, scikit-learn

# Cost to Performance - Resource Usage - AI

Resource usage by AI based Anomaly Detection Script on Pi5

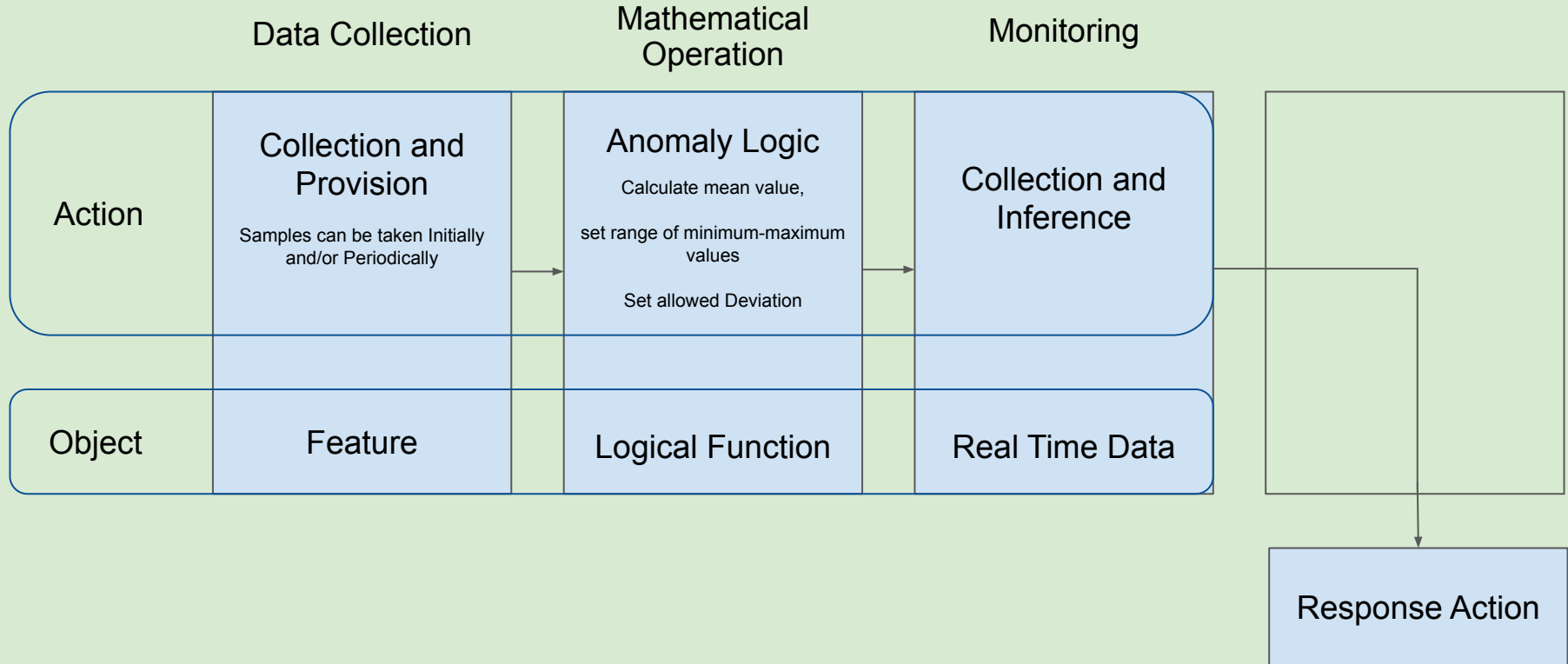
**CPU utilization:** ~3.5% of one logical CPU (~0.87% of total 4-core capacity)

**Memory Footprint (RSS) :** ~3.7% / ~160 MiB

Above Demonstrative Data are very general rough estimation.

For actual impact, these and other performance indicator value required to be derived from multiple parameter reading by running application

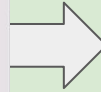
# Architectural Model - Embedded based Solution for Microcontroller



# Proof of Concept Execution - Logic based

## Data Collection

```
bonny-p@Nebula: ~/Documents/Projects/EAG_IDS/Logic-based_C_Program
File Edit Tabs Help
bonny-p@Nebula:~/Documents/Projects/EAG_IDS/Logic-based_C_Program $ ./Data_Collector.exe
=== C Data Collector ===
Config      : config.json
Output      : system_data.csv
Interval    : 2 s
CPU         : ON
RAM         : ON
Process count: ON
[2026-09-21 11:44:05] CPU= 0.00% RAM= 64.96% Processes=694
[2026-09-21 11:44:08] CPU= 2.55% RAM= 65.04% Processes=695
[2026-09-21 11:44:10] CPU= 2.44% RAM= 65.16% Processes=696
[2026-09-21 11:44:12] CPU= 4.91% RAM= 65.15% Processes=698
[2026-09-21 11:44:14] CPU= 2.78% RAM= 65.05% Processes=699
[2026-09-21 11:44:16] CPU= 2.20% RAM= 65.04% Processes=697
[2026-09-21 11:44:19] CPU= 3.52% RAM= 64.97% Processes=697
[2026-09-21 11:44:21] CPU= 2.09% RAM= 64.95% Processes=697
[2026-09-21 11:44:23] CPU= 3.13% RAM= 64.99% Processes=697
[2026-09-21 11:44:25] CPU= 2.73% RAM= 65.00% Processes=697
[2026-09-21 11:44:27] CPU= 2.43% RAM= 64.98% Processes=696
[2026-09-21 11:44:29] CPU= 1.97% RAM= 64.88% Processes=695
[2026-09-21 11:44:32] CPU= 2.85% RAM= 64.93% Processes=694
```



## Logic based generated Output - Detection

```
bonny-p@Nebula: ~/Documents/Projects/EAG_IDS/Logic-based_C_Program
File Edit Tabs Help
bonny-p@Nebula:~/Documents/Projects/EAG_IDS/Logic-based_C_Program $ ./Monitor.exe
=====
C Statistical Anomaly Monitor
No AI / No Machine Learning
=====
Input : system_data.csv
Output: analysis_results.csv

=== Calculated Baseline ===
CPU      : mean=2.58 min=0.00 max=4.91 stddev=1.05 range=[-0.55, 5.72]
RAM      : mean=65.01 min=64.88 max=65.16 stddev=0.08 range=[64.78, 65.24]
Processes : mean=696.31 min=694.00 max=699.00 stddev=1.43 range=[692.00, 700.61]

=== Runtime Analysis ===
[2026-09-21 11:44:05] CPU=NORMAL RAM=NORMAL Processes=NORMAL => NORMAL
[2026-09-21 11:44:08] CPU=NORMAL RAM=NORMAL Processes=NORMAL => NORMAL
[2026-09-21 11:44:10] CPU=NORMAL RAM=NORMAL Processes=NORMAL => NORMAL
[2026-09-21 11:44:12] CPU=NORMAL RAM=NORMAL Processes=NORMAL => NORMAL
[2026-09-21 11:44:14] CPU=NORMAL RAM=NORMAL Processes=NORMAL => NORMAL
[2026-09-21 11:44:16] CPU=NORMAL RAM=NORMAL Processes=NORMAL => NORMAL
[2026-09-21 11:44:19] CPU=NORMAL RAM=NORMAL Processes=NORMAL => NORMAL
[2026-09-21 11:44:21] CPU=NORMAL RAM=NORMAL Processes=NORMAL => NORMAL
[2026-09-21 11:44:23] CPU=NORMAL RAM=NORMAL Processes=NORMAL => NORMAL
[2026-09-21 11:44:25] CPU=NORMAL RAM=NORMAL Processes=NORMAL => NORMAL
[2026-09-21 11:44:27] CPU=NORMAL RAM=NORMAL Processes=NORMAL => NORMAL
[2026-09-21 11:44:29] CPU=NORMAL RAM=NORMAL Processes=NORMAL => NORMAL
[2026-09-21 11:44:32] CPU=NORMAL RAM=NORMAL Processes=NORMAL => NORMAL
```

C based executable available @ [https://github.com/bonny-p/Embedded\\_Anomaly\\_Guard](https://github.com/bonny-p/Embedded_Anomaly_Guard)

Platform: RTOS, Linux, Windows, MacOS

Dependencies: pip - psutils, pandas, scikit-learn

# Cost to Performance - Resource Usage - Embedded Logic

Resource usage by AI based Anomaly Detection Program on Pi5

**CPU utilization:** ~ <1%

**Memory Footprint (RSS) :** ~0.03% / ~2 MiB

Above Demonstrative Data are very general rough estimation.

For actual impact, these and other performance indicator value required to be derived from multiple parameter reading by running application

# AI vs Embedded solution

1-5 Parameters for Analysis → Embedded Solution is sufficient and beneficial.

6-10 Parameters for Analysis → Use case based solution is required to be evaluated

10+ parameters for Analysis → AI solution is required to identify anomalies with all possible combination values.

AI based exclusive detection - Use case

Example: All Parameters values are within range but combination can identify the Anomaly

- **Consistent/Normal:** High Traffic or High number of running Processes & High CPU/RAM Usage
- **Anomaly:** Low Traffic or Low number of running Processes & High CPU/RAM Usage